# POPLMark goes Concurrent! Challenge problems for process calculi and behavioural types

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**Abstract**

The publication of POPLMark spearheaded the era of publishing formal proofs alongside with new research. This fostered an era of high-quality and high-assurance proofs at top venues for programming language research. This paper simultaneously acknowledges the impact of the original POPLMark challenge, and proposes a new benchmark intended to stimulate the research and dissemination of techniques that address the particular challenges of process calculi and behavioural type systems. In this work we propose three challenge problems concerning to process calculi and name passing, linearity in behavioural type systems, and coinductive reasoning for process algebras.

## 1 Introduction

In recent years, the programming language community has embraced machine checked proofs for theoretical results in papers. The POPLMark Aydemir et al. [2005] challenge was proposed for this purpose, and to get the community on track. Since then the number of formal proofs accompanying papers has grown[1]. However, formal proofs are not equally spread in the field. In this call to action, we propose to extend the key contribution of challenges like POPLMark is to help develop the necessary momentum and culture to be able to formalise results to the concurrent calculi community by proposing a challenge the exercises the key aspects of reasoning about process calculi. We propose this document, and a website to enable the community to collaborate on the project.

It has been a long time since the original challenge, and many changes have happened. Particularly, POPLMark and any challenge cannot address every conceivable technique. Challenges have to be simple so that the techniques are not shadowed by technical details and also feasible to implement without heroic efforts. In that vein, it is apparent that new challenges are needed, work like POPLMark Reloaded Abel et al. [2019] strives to extend the scope of the

[1] add some POPL/PLDI/etc statistics here

original challenge to proofs using logical relations. This technique is crucial to many current developments.

As mentioned before, in this work we address another expansion of scope for such challenges, going from the $\lambda$-calculus to process calculi such as the $\pi$-calculus and session type systems (and other behavioural type systems). These systems require particular insights to implement. So far, we have identified three key aspects[2]. First, the idea of scope extrusion in name passing calculi. This generalises the idea of binders in ways that techniques that were adequate for $\lambda$-calculus binders are not so convenient for name passing systems. Second, session types in particular, and behavioural types in general often depend on linearity in their type systems. The formalisation of linear type systems posses a challenge given that it is not uniformly well supported among the many proof assistants. Finally, processes in concurrent systems often contain infinite behaviour that is typically modelled with coinduction. This technique is used to reason about (bi)similarity between processes, the equivalence of their traces, or subtyping between infinite trees of possible behaviours. This challenge concentrates on those issues.

In this work we propose a challenge where each part independently exercises one of the three areas, however it is often the case that a problem requires more than one of these techniques simultaneously. In this work, we explore them in isolation, in order to facilitate the implementation of smaller solutions. Concretely, we do not address the idea of how to combine these techniques, leaving the discussion of that to the future when we have collected enough entries to these problems.

We do not claim in this work that these problems have never been approached and formalised before. In fact, the contrary is true. All of these problems have been addressed in several forms. This motivates this challenge to enable an easier comparison between solutions (since they all implement the same challenge), and in a setting that is *simple enough to not be distracting, yet complex enough that we can compare strengths and weaknesses of each.*

The rest of this paper is structured in the following way: section 2 expands on the rationale for the selected challenge problems. Then section 3 presents the problems. We discuss the solutions in section 4, in particular section 4.1 walks through a tutorial solution. And finally section 5 presents some conclusions and it contains a call to action to invite the submission of new solutions.

## 2 Rationale for choosing the problems

This section is not (at least initially) meant to be in the final manuscript but it is to document some reasoning for choosing appropriate problems. The idea is to explore three areas: scope extrusion in name passing calculi, linearity requirements in behavioural types, and finally coinductive reasoning in trace equivalence, (bi)similarity, or subtyping in the presence of recursion.

[2]We want to keep the number low, but the exact nature of the challenges is still open to discussion (as are most things in this project).

The idea is to have three small problems in the aforementioned subjects that are independent of each other and that are easy to understand and to implement. This is important so we have more entries. They have to be small so it is not a lot to read, and independent so people may choose to implement only one of them, or work on them in any order. Moreover, the specification should only need to be met in spirit. The idea is to provide a solution author with some space to show the best aspects of their solution even if that means adapting the problem definition while keeping the spirit of the challenge.

Finally, it will be important to mention in the rationale for problems that the key aspects are:

- Scope extrusion
- Linearity
- Coinduction

But the exact presentation of the concrete problems should be somewhat flexible. In the sense that the use of certain techniques or tools may require changing the presentation of the system to fit the technique. These changes should be allowed, and even encouraged, but the result should be "morally" the same system and there should be a rationale for the changes in presentation. That way we could not only see valid techniques but also what presentation of the rules to use and the rationale to chose a particular approach.

# 3 The challenge problems

The big picture for the benchmark can be seen in fig. 1 where the three pillars (i.e.: concepts typically needed for process calculus proofs) appear in blue, with some ideas of what to prove. And a second step of combining the three aspects (for a total of six steps for the whole benchmark), and of course the +1-step for the combination of everything on the reader's next research papers.

## 3.1 Name passing and scope extrusion

**Problem idea**: A small proof about the untyped $\pi$-calculus.

## 3.2 Linearity and behavioural type systems

**Problem idea**: A small session type problem that exposes linear channels.

## 3.3 Coinduction and reasoning about process algebras

**Problem idea**: Something about traces, (bi)similarity, or subtyping under non terminating problems in behavioural types.

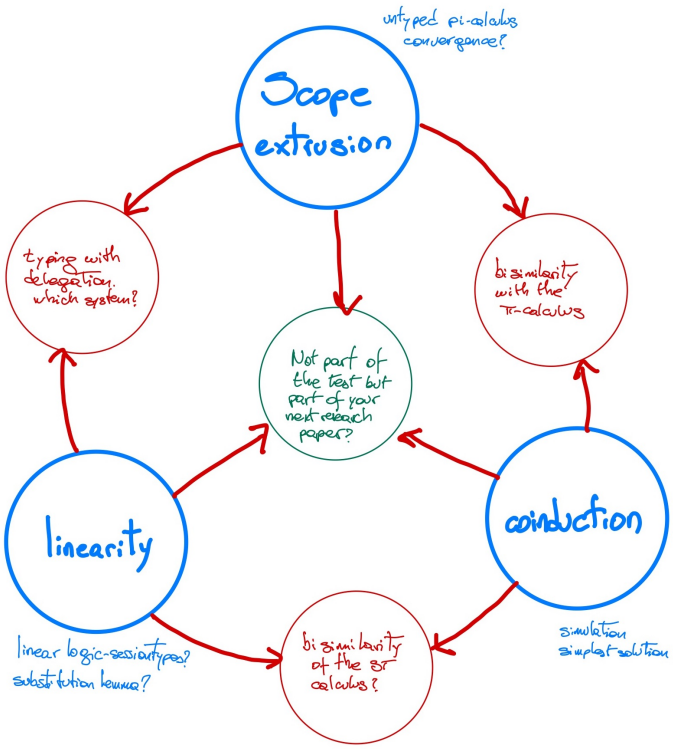


Figure 1: The benchmark's big picture.

# 4 The solutions

## 4.1 A tutorial solution on paper

Probably the skeleton here and more details on an appendix.

## 4.2 Flexibility in solving a challenge

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# 5 Conclusion and related work

# References

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